

Reel Connections



DATA 515 Final Project • Owen Guo, Zach Lubarsky, Henry Shi, Daniel Yan • University of Washington

PROJECT OVERVIEW

What We Built

An interactive Streamlit web app that challenges users to connect movies and actors — inspired by the classic parlor game Six Degrees of Kevin Bacon.



Front End & Backend

Streamlit UI connecting to Python code



Data Sources

Leverages TMDb datasets to map real actor-movie relationships, Box Office totals from *WatchMojo*.



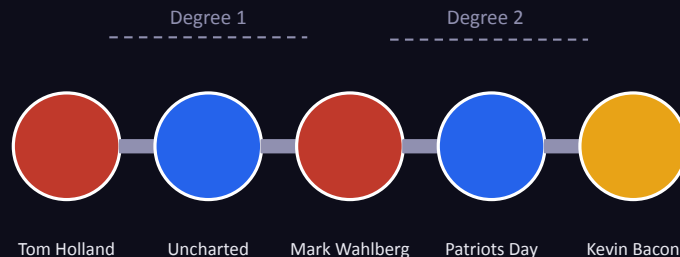
CI/CD Principles

Use CI/CD, PyLint, and Unit Tests to keep code quality high.

THE GAME CONCEPT

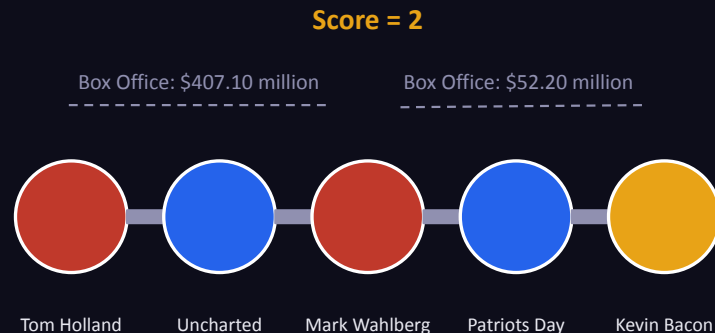
Game Mode 1

Connect the actors in the *minimum* number of stops

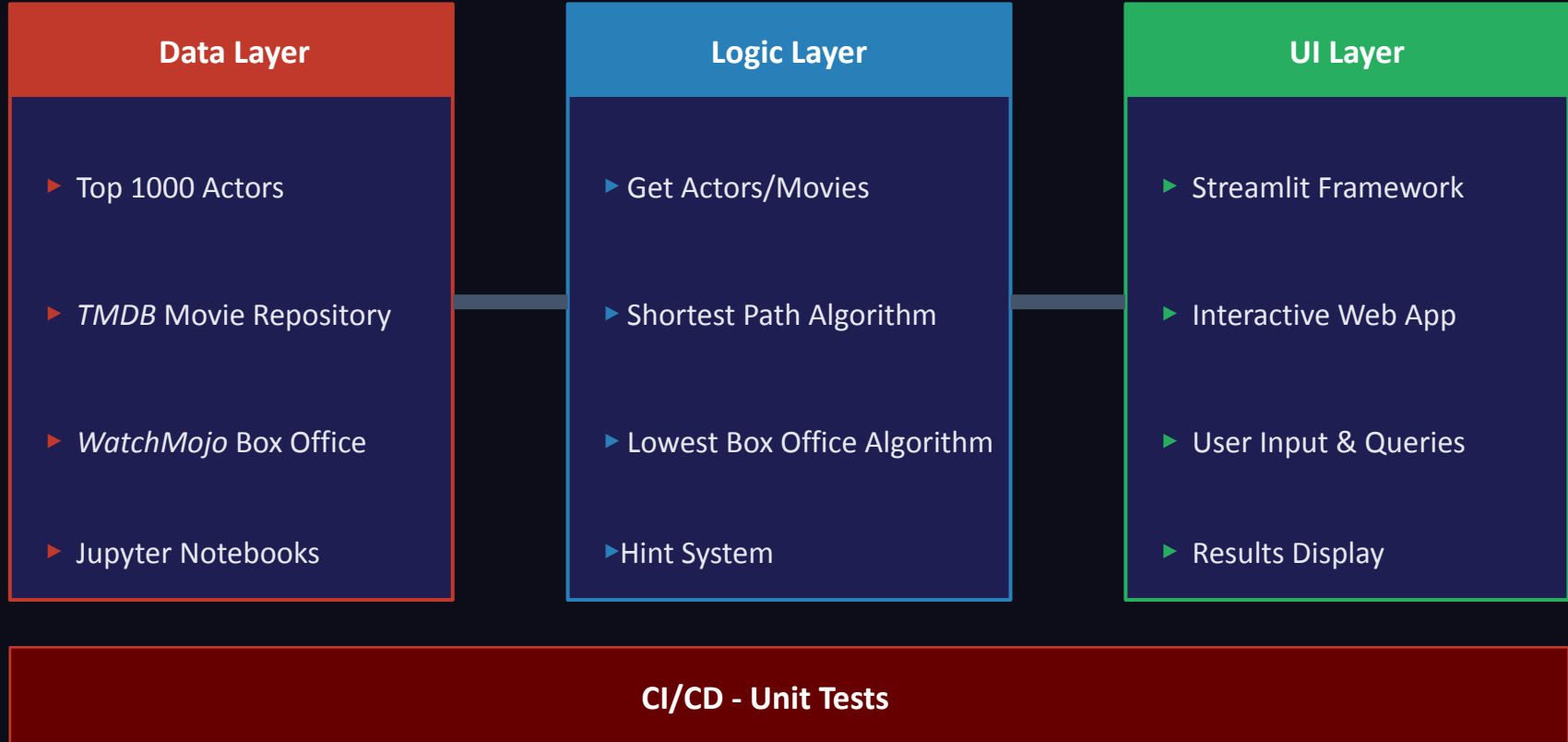


Game Mode 2

Connect the actors in the *lowest box office*



TECHNICAL ARCHITECTURE



Movie Actor Link Game

▼ 🎮 How to Play

Connect the **Start Actor** to the **Target Actor** by hopping through movies and co-stars.

How to play

1. Start from the current actor.
2. Choose a movie that the actor appeared in.
3. Pick a co-star from that movie to become your next actor.
4. Repeat until you reach the target actor.

Game modes

- **Normal Mode:** win in as **few steps** as possible (each move counts as 1 step).
- **Challenge Mode:** win with the **lowest total box office** (each chosen movie adds its box office to your total).

🏆 **Winning** You win as soon as your current actor matches the target actor. Your score is compared to the algorithm's optimal path.

Normal Mode

Challenge Mode

Demo



Normal Mode

>  Hint

Start Actor

Craig Robinson

Target Actor

Nicholas Hoult



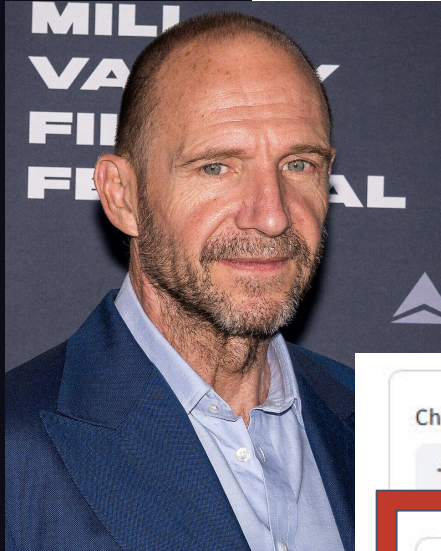
Selected Movie: Dolittle (2020)

Next Actor (type to search, or open the menu to select)

Dolittle

Confirm Next Actor

Demo



Normal Mode

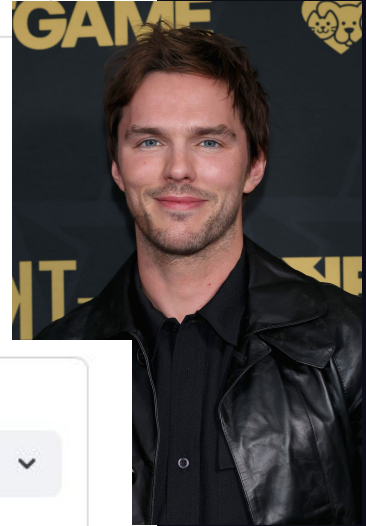
> 💡 Hint

Start Actor

Ralph Fiennes

Target Actor

Nicholas Hoult



Choose a Movie

The Menu (2022)

Confirm Movie

Restart

Back to Home

Demo

Result

🎉 You connected to the target actor!

Steps Used: 5

Optimal Steps: 2

Score: 40.0

You can do better next time!

Your Path

Keanu Reeves → Good Fortune (2025) → Seth Rogen

Seth Rogen → Knocked Up (2007) → Leslie Mann

Leslie Mann → The 40 Year Old Virgin (2005) → Steve Carell

Steve Carell → Anchorman: The Legend of Ron Burgundy (2004) → Will Ferrell

Will Ferrell → The Goods: Live Hard, Sell Hard (2009) → Craig Robinson

Optimal Path

Keanu Reeves → Good Fortune (2025) → Seth Rogen

Seth Rogen → Knocked Up (2007) → Craig Robinson

Demo

<https://reel-connections.streamlit.app/>

Questions?

KEY TAKEAWAYS

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01

Graph theory provides a powerful framework for analyzing real-world social networks like Hollywood actor relationships.

02

Streamlit enables rapid deployment of data-driven apps that are accessible to non-technical users.

03

IMDb datasets are rich enough to support meaningful media analysis and interactive exploration.

04

Gamification increases engagement with data — turning analysis into an interactive experience.